

Sai Madhav Rayabandi

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Work Authorization: Legally authorized to work in the U.S. without sponsorship.

Professional Summary

- Full-stack developer with **5+ years of experience** building and maintaining production-grade web applications across healthcare, AI, and enterprise sectors.
- Strong backend expertise in **Java (Spring Boot)**, **Python (Django)**, **.NET Core**, and **Node.js**, following OOP, SOLID, and scalable microservices architecture.
- Built responsive front-end applications using **React.js**, **Angular**, and **Vue.js**, creating reusable components and dynamic single-page interfaces with modern JavaScript and TypeScript.
- Designed and deployed intelligent **LLM-based chatbots** and **Retrieval-Augmented Generation (RAG)** systems using **LangChain**, **Ollama**, **Gemini**, **AWS Bedrock**, and **SageMaker**.
- Skilled in **UI/UX design** using **Figma**, developing wireframes, mockups, and delivering pixel-perfect frontend implementations aligned with user experience best practices.
- Hands-on experience with **PostgreSQL**, **MySQL**, **MongoDB**, and **Snowflake**, with expertise in schema design, indexing, joins, and query optimization.
- Deployed cloud-native applications using **AWS** (EC2, S3, Lambda, ECS, Fargate, MSK) and managed MLOps workloads on **SageMaker**, including model training and hosting.
- Built and deployed microservices using **Docker**, with fully automated CI/CD pipelines powered by **GitHub Actions**, **Jenkins**, and **Octopus Deploy** on AWS.
- Developed and integrated secure **RESTful APIs**, implemented **JWT-based authentication**, and enabled OAuth2 flows for secure communication across services.
- Experience working with GraphQL APIs and resolvers in Node.js applications for efficient frontend-backend integration.
- Familiar with AWS DynamoDB and AppSync architecture for serverless GraphQL-based data access and real-time updates.

Technical Skills

- **Languages:** Java, Kotlin, Python, C#, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3.
- **Frameworks and Libraries:** Spring Boot, Django, .NET Core, React.js, Angular, Vue.js, Node.js, Express.js, Bootstrap, Material UI.
- **Cloud and DevOps:** AWS (EC2, S3, Lambda, SageMaker, Bedrock), Azure, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, Octopus Deploy.
- **Databases:** PostgreSQL, MySQL, MongoDB, SQLite, Snowflake, Oracle, Firebase Realtime/Firestore.
- **Machine Learning / LLM:** LangChain, Ollama, Gemini, Claude, AWS Bedrock, OpenAI, BERT, Transformers, Retrieval-Augmented Generation (RAG), HuggingFace.
- **API Development:** RESTful APIs, GraphQL, JWT Authentication, OAuth2, Swagger/OpenAPI documentation.
- **Testing QA:** JUnit, Mockito, pytest, Selenium, Jest, Mocha, Postman.
- **UI/UX Design Tools:** Figma, Adobe XD, Responsive Web Design, Accessibility (WCAG), Cross-Browser Compatibility.
- **Version Control Agile Tools:** Git, GitHub, Bitbucket, JIRA, Confluence, Agile/Scrum methodologies.

Experience

Software Developer

January 2023 –Present

University of Nebraska at Omaha

- Designed and deployed **three Django-based healthcare platforms** with responsive UIs, built using HTML, CSS, and JavaScript, tailored to research teams and medical professionals.
- Created custom UI layouts from scratch, improving navigation and accessibility across desktop and mobile devices; integrated design feedback using **Figma wireframes**.
- Built a **fully functional AI-powered chatbot** using **Google Gemini API**, converted to a **Retrieval-Augmented Generation (RAG)** system using LangChain and Ollama to pull answers strictly from published research papers stored in AWS S3.
- Managed **CI/CD deployment pipelines** using GitHub Actions and AWS CodePipeline to automate testing, containerization (Docker), and deployment on EC2 and Lambda.
- Developed and evaluated multiple **LLM integrations** using LangChain, comparing output consistency, accuracy, and response time for different models (Gemma, Claude, Mistral, Phi, Deepseek).
- Created the **SLP Core Tool** — a Java-based machine learning assistant that predicts the next code token with up to **90% accuracy** by analyzing user input and parsing abstract syntax trees from StackOverflow-trained data.
- Preprocessed and filtered **100GB+ of StackOverflow data** to extract high-quality code samples for fine-tuning and benchmarking the SLP model.
- Documented technical workflows, API structures, and user guides; regularly demonstrated deliverables to stakeholders and academic supervisors for iterative feedback.

Web Developer

May 2020 –December 2022

Infosys Pvt. Ltd.

- Worked on two enterprise-scale applications — a travel guide app (**Wonderlust**) and an **e-commerce platform** for Siemens Energy.
- Developed RESTful APIs using **Java Spring Boot**, implementing controllers and business logic for login, validation, and user flow in Wonderlust.
- Created reusable UI components using **React** such as tourist destination cards with dynamic content, enhancing the frontend dashboard.
- For Siemens Energy, worked on SAP Hybris-based e-commerce platform, designing category-specific pages and page layouts using templating systems.
- Enhanced the **Add to Cart** feature by developing dynamic components and improving flow handling for large product inventories.
- Integrated Kafka-based event pipeline to handle order-based email triggers and user notifications, ensuring delivery and tracking.
- Managed code build and deployment to **development and production environments**, tracked logs, errors, and server metrics using **Kibana**.
- Wrote optimized SQL queries in **PostgreSQL** and secured backend APIs using OAuth and JWT for session handling.
- Followed Agile development practices, managed Git repositories via SourceTree, and participated in sprint reviews, defect triage, and feature releases.

Projects

Weather Forecasting System (Django + ML)

- Built a Django-based application to predict next-day weather using linear regression on Kaggle datasets with inputs like temperature, humidity, and wind speed.
- Designed responsive UI using HTML, CSS, Bootstrap, and JavaScript to visualize forecast output dynamically.

Kinematics Solver (Django + ML)

- Developed an AI tutor-like web app to solve physics kinematics problems using keyword extraction and formula matching via Django ML pipeline.
- Built clean, responsive frontend with Bootstrap and JavaScript for input parsing and formatted answer explanation.

Fake News Detection App (OCR + NLP)

- Created a web-based system allowing users to upload news screenshots; extracted text using Tesseract OCR and classified using a custom-trained NLP model.
- Displayed predictions with probability scores and supported batch analysis via file uploads.

Budget Tracker App (Node.js + React)

- Developed full-stack budgeting app where users could log daily purchases and visualize categorized monthly spending trends using charts.
- Built secure backend with Node.js and JWT; integrated dynamic React.js components and deployed responsive UI with charting libraries.

Multi-LLM RAG Testing Platform (Django + LangChain)

- Created an internal Django platform to evaluate and compare multiple fine-tuned LLMs using Retrieval-Augmented Generation with documents from AWS S3.
- Integrated LangChain and Ollama to run inference, fuzzy match FAQs, and visualize performance side-by-side.

Health and Beauty AI Assistant (Django + OpenAI)

- Developed a user-friendly AI assistant to provide personalized skin, hair, and wellness suggestions based on user input through structured follow-up forms.
- Built elegant UI in HTML, CSS, Bootstrap with interactive logic in JavaScript; integrated LLM API for natural responses.

Certifications

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| • Problem Solving — HackerRank Certification | Certified |
| • JavaScript — HackerRank Certification | Certified |
| • Cloud Developer (in progress) — AWS Academy | Ongoing |
| • Figma for UI/UX Design — Simplilearn Certification | Certified |
| • Cascade CMS Content Management — University of Nebraska at Omaha | Certified |